Workstream Architecture Brief

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Governed contribution infrastructure for human and AI work

Workstream turns project-defined tasks, immutable submissions, policy-governed checks, and authorized review into trusted `ContributionRecord` facts. It records who completed what, under which locked rules, using which exact artifact, and with what verified outcome.

Scope: bounded v0.1 delivery, with future adapter context for identity, task contracts, settlement, and reputation.

Executive Summary

Workstream is the governed lifecycle core between systems that request work and systems that consume its outcome. It does not need to own the source application, execution environment, identity provider, or downstream economic system. It gives every project versioned rules, every task a locked policy context, every Submission immutable artifact lineage, every valid human decision an attributable Review and reviewer contribution, and every accepted task an immutable FinalAcceptance before the submitter contribution.

v0.1 is focused on proving the internal lifecycle:

```
Project Guide -> Task Queue -> Pre-Submission Intake -> Immutable Submission
-> Post-Submission Work Evaluation -> Review
-> Revision / FinalAcceptance / Rejection -> Contribution Record
-> Compensation Award / Fulfillment -> deferred reputation projection
```

Pre-submission intake checks package fitness, completeness, integrity, and configured intake quality before Submission creation. Post-submission evaluation checks the immutable work against locked task/project requirements and produces durable review-eligibility evidence. Passing intake is not proof of task success; neither stage replaces authorized Review.

The stages are not a blanket guarantee of deterministic execution. Deterministic rules and supported model/agent evaluators are distinct methods; deterministic policy compilation does not guarantee repeatable model judgments. Only supported registered implementations execute. The setup agent proposes policies, not runtime judgments, and no live agent judge is claimed by this brief. The roadmap owns current implementation status.

> Current v0.1 is backend-first and internal-loop-first. External source adapters, agent identity writes, task escrow, x402 payment requests, OmniClaw settlement, USDC payouts, public marketplace flows, and automated routing remain adapter boundaries until the internal evaluation loop works with real tasks.

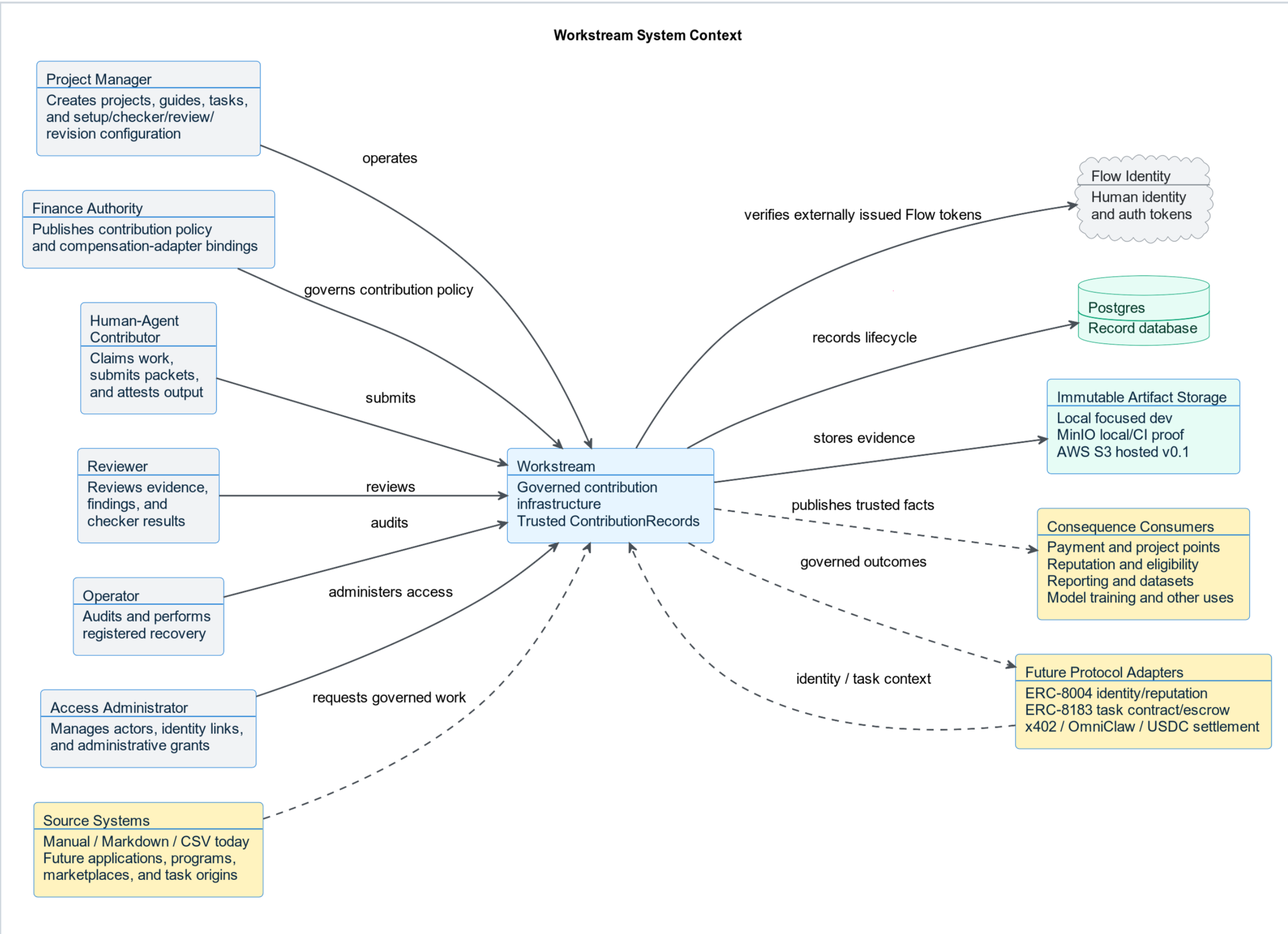
Architecture Principles

Principle	Meaning
Source-agnostic, manual-first	v0.1 accepts manual, markdown, or CSV-controlled intake. Future origins normalize into the same task contract.
Identity is not authority	Flow Identity is the current v0.1 authentication provider. Local grants and lifecycle guards decide Workstream authority.
Trusted contribution facts	Exact artifact, locked policy, checker, Review, and actor lineage produce immutable `ContributionRecord` facts for downstream consumers.
Modular monolith	FastAPI remains one deployable backend while keeping routers, services, repositories, ports, and adapters separate.

Principle	Meaning
Postgres record database	Local, CI, and production-like development use Postgres as the record database.
Object-storage abstraction	Local filesystem storage is allowed only behind the provider-neutral `ArtifactStore`; AWS S3 is the v0.1 hosted provider and MinIO is the local/CI protocol proof.
Async-first execution	Long-running checker work does not block request/response paths.
Contribution before compensation	Every valid human Review creates a reviewer contribution. FinalAcceptance is created only for accept and is the sole source of the submitter contribution. Compensation may attach afterward; reputation is deferred.

C1: System Context

The context diagram shows Workstream between source systems and consequence consumers. Workstream owns governed lifecycle truth. It does not own primary identity, work execution, external task origins, settlement rails, reputation systems, or other uses of the resulting contribution facts.

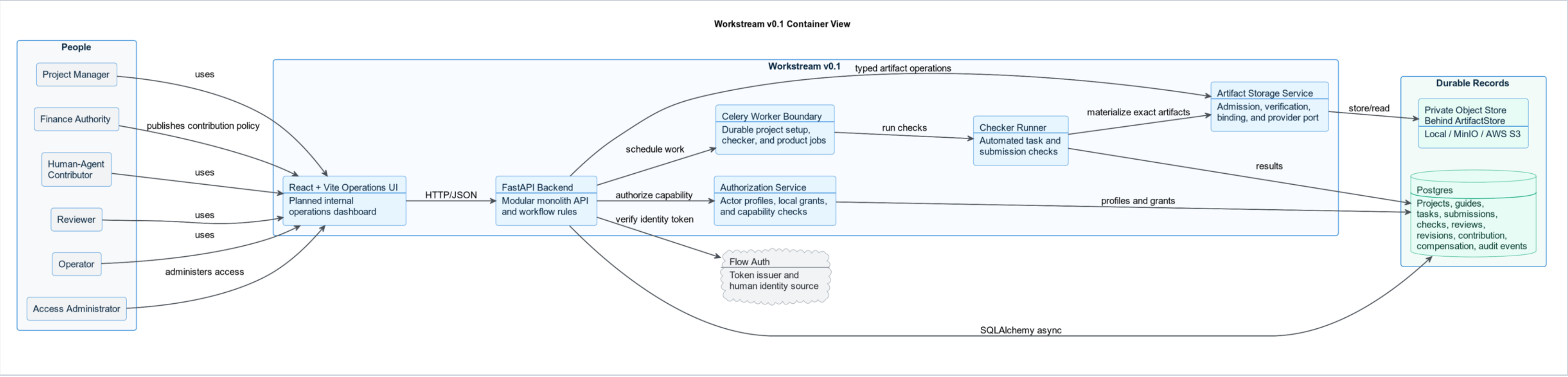


What This Means

- Project managers, contributors, reviewers, operators, Finance Authorities, Access Administrators, and Audit Authorities interact with Workstream through their independent grants.
- Adjudicator grants are independent but authorize no v0.1 lifecycle or action.
- Flow Identity is the current v0.1 human identity and authentication source, not the definition or ownership boundary of Workstream.
- Postgres is the record database.
- Storage sits behind an object-storage abstraction.
- Source applications and future protocol rails connect through adapters.
- Payment, points, reputation, reporting, datasets, and model-training systems consume Workstream facts without creating or rewriting them.

C2: v0.1 Container View

The container view shows the bounded v0.1 implementation. It is intentionally small: React + Vite for the planned internal operations UI, FastAPI for the backend, Postgres for records, a storage interface for artifacts, and an async checker/job boundary.

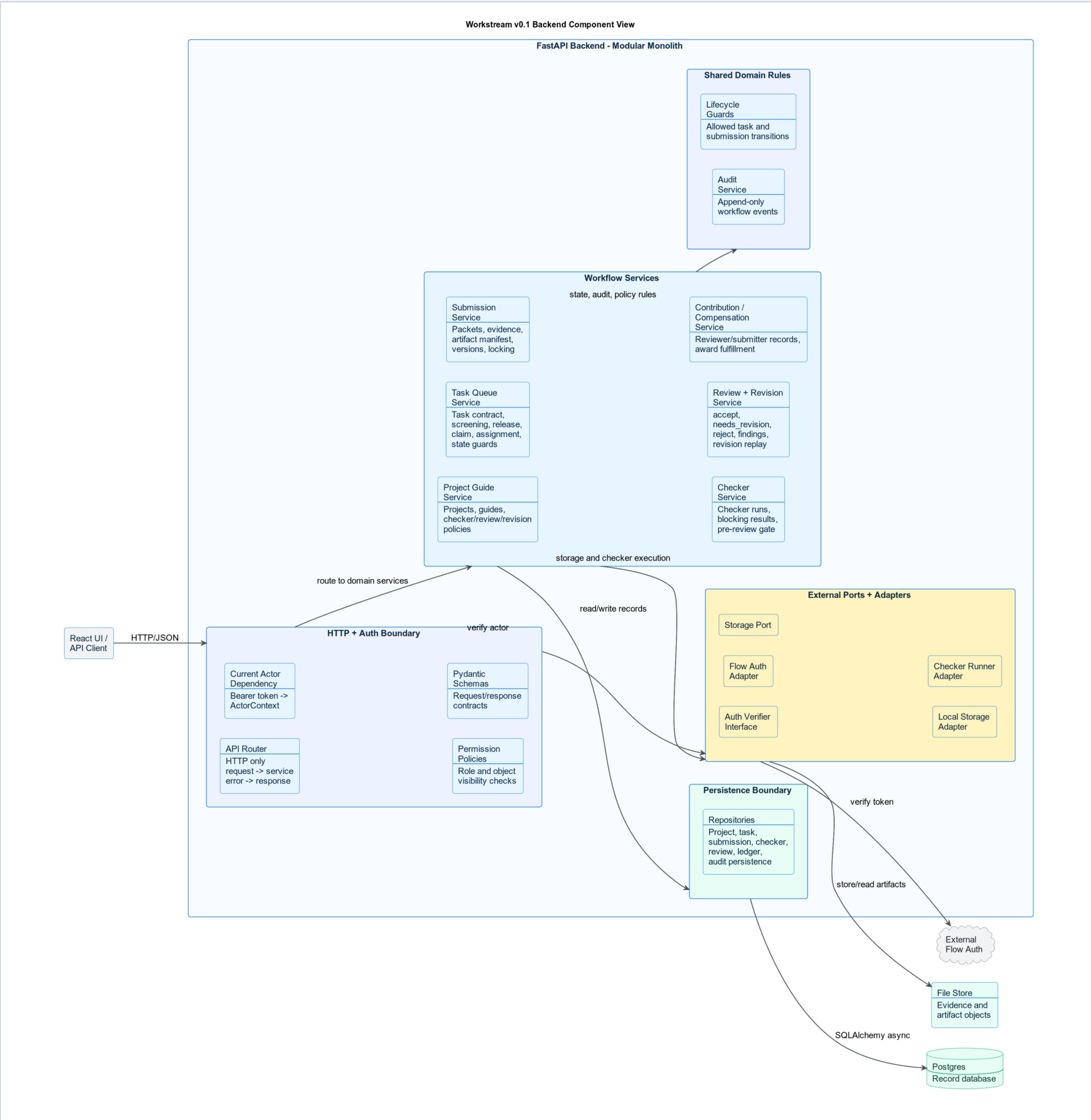


Container Responsibilities

Container	Responsibility
React + Vite operations UI	Planned internal operations dashboard for project, task, submission, review, and compensation fulfillment workflows. Reputation UI remains deferred.
FastAPI backend	API contracts, workflow rules, auth dependency, lifecycle guards, module orchestration, and audit writes.
Celery worker boundary	Durable project setup, checker, and background product-job execution. FastAPI background tasks are not the Workstream product-job boundary.
Checker runner	Executes automated checks and stores checker results.
Storage interface	Keeps file/evidence semantics stable while local storage and the hosted AWS S3 profile implement the same provider-neutral port.
Postgres	Durable record database for the full Workstream lifecycle.

C3: Backend Component View

The backend component view zooms into the FastAPI container. It shows how the modular monolith stays clean without becoming a distributed system too early.

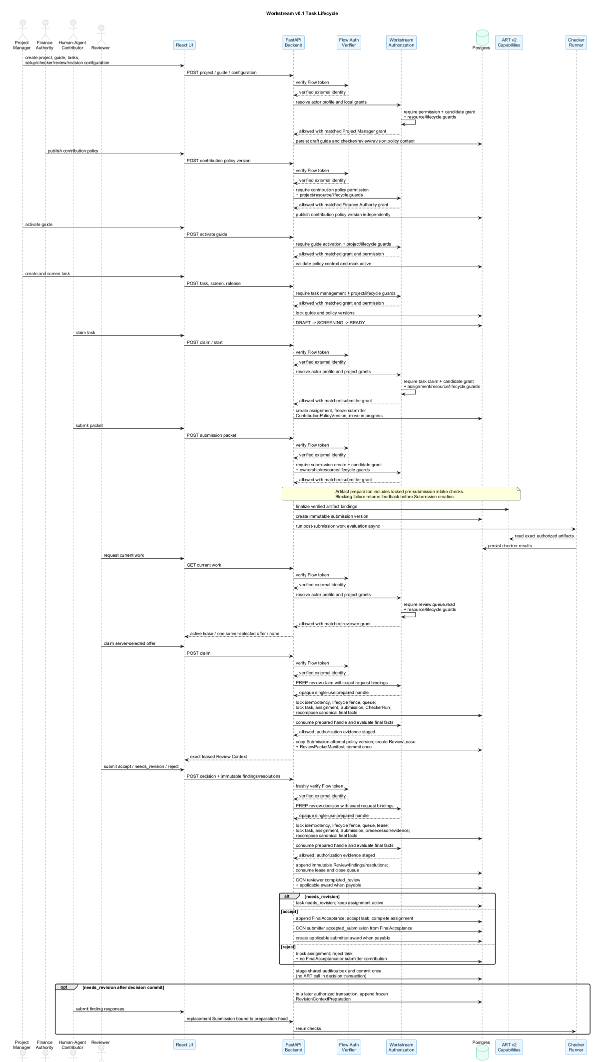


Backend Boundaries

Boundary	Responsibility
HTTP + auth boundary	Routers handle HTTP only. Actor resolution, permission checks, and Pydantic request/response validation stay at the boundary.
Workflow services	Project guide, task queue, submission, checker, review/revision, and contribution/compensation services own business rules; reputation remains deferred.
Shared domain rules	Lifecycle guards and audit writes stay shared instead of being scattered through routers.
Persistence boundary	Repositories own SQLAlchemy async persistence and Postgres access.
External ports/adapters	Flow auth, storage, and checker execution stay behind interfaces.

Lifecycle Sequence

The sequence below shows the narrow v0.1 loop the system must prove before expansion.

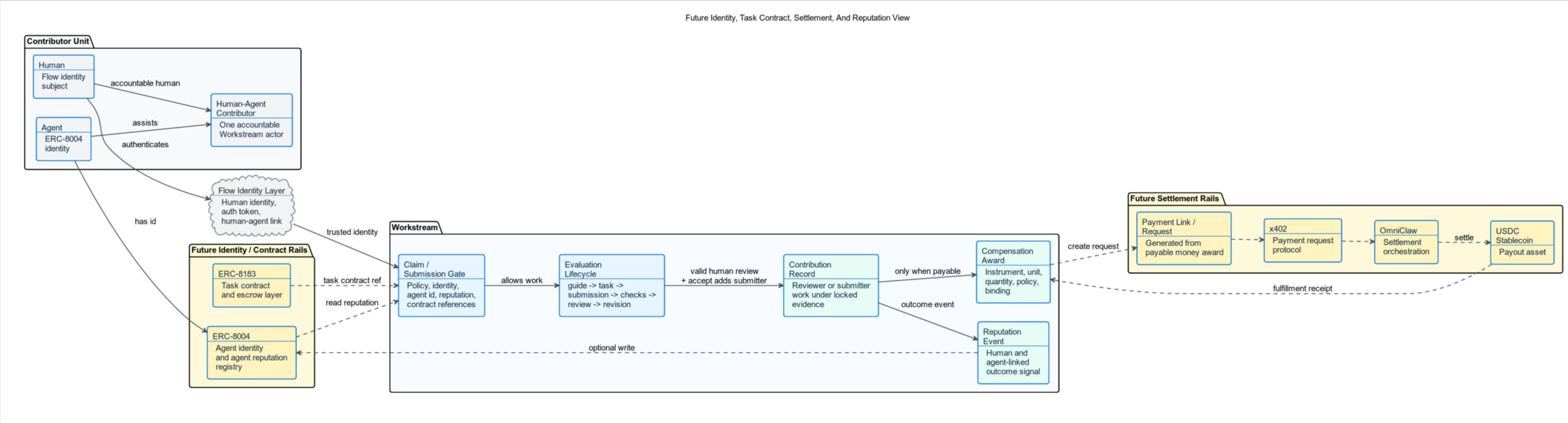


Lifecycle Invariants

- A task cannot enter `READY` until it has locked the complete, binding-valid ContributionPolicyVersion bound during guide activation. TaskAssignment copies that task lock, Submission stamps the exact attempt version, and ReviewLease copies the Submission stamp without claim-time selection. A version retired after it was validly locked remains authoritative for that immutable attempt.
- A contributor submission creates a new immutable submission version; locked artifacts are not edited in place.
- Review decisions are exactly `accept`, `needs_revision`, or `reject`.
- `needs_revision` preserves immutable findings, responses, preparations, and later resolutions.
- Every valid human Review creates a reviewer contribution; accept additionally creates FinalAcceptance, which alone sources the submitter contribution.
- Compensation fulfillment status is separate from task acceptance.

Future Identity, Task Contract, Settlement, And Reputation

This view explains the broader architecture direction without moving it into v0.1 scope.



Future Separation Of Concern

Concern	Owner
Current v0.1 human identity and auth	Flow Identity adapter
Agent identity	ERC-8004
Agent reputation read/write	ERC-8004 through a future Workstream adapter
Task contract and escrow reference	ERC-8183
Governed task, artifact, check, review, and revision lifecycle	Workstream
Reviewer and accepted-submitter contribution facts	Workstream `ContributionRecord`
Contribution policy, immutable award, and fulfillment status	Workstream compensation records
Payment request and settlement execution	x402, OmniClaw, and USDC settlement rails

> Future ERC-8004, ERC-8183, x402, OmniClaw, and USDC integrations do not replace Workstream. They use Workstream records. Source and consequence systems cannot create or revise Workstream identity, authority, submission, Review, or contribution truth.

Scope Boundary

Current v0.1 Boundary

- project guide and versioned policy context
- task queue and task records
- assignment and claim flow
- submission packets and evidence
- checker framework and pre-review gate
- human review and revision replay
- contribution records
- compensation award, receipt, and fulfillment projection records
- contribution evidence for a future reputation projection
- audit events

Later Adapter Boundaries

- external origin onboarding and source adapters
- automated routing
- owner-agent execution workspace
- ERC-8004 agent identity and reputation writes
- ERC-8183 task contract and escrow settlement
- x402 payment requests
- OmniClaw settlement orchestration
- USDC payout execution
- marketplace discovery
- adjudication lifecycle, actions, queues, leases, and decisions

Closing

Workstream v0.1 succeeds when it can run real internal work from project guide to reviewer/submitter contributions with evidence, checks, immutable human review, revision discipline, FinalAcceptance, conditional compensation awards, and fulfillment status. The active review contract is `docs/spec_review_lifecycle.md`; its surfaces remain unavailable until REV-13.

The system should expand only after that loop is proven.

The complete product remains source-agnostic: once the governed core is proven, centralized, sovereign, federated, and permissionless applications can consume the same trusted contribution facts through explicit adapters.